SYSDESIGN REFERENCE

Read and Write Quorums

A senior-level reference with tradeoffs, failure modes, and interview follow-ups for read and write quorums.

Use overlapping read and write sets to tune consistency and availability in replicated storage.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / QUORUMS

SYSDESIGN

Read and Write Quorums

CORE IDEA

Use overlapping read and write sets to tune consistency and availability in replicated storage.

Reference architecture decision chain

01	02	03	04
N/R/W Values	Sloppy Quorums	Conflict Resolution	Repair

Decisions to defend

+ N/R/W values
+ sloppy quorums
+ conflict resolution
+ repair
+ and latency.

Failure modes to test

+ Assuming quorum guarantees recency without versions and ignoring correlated failures.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Read and Write Quorums

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.